

WEEK – 8: Interfaces in Java — Employee Salary Details

Experiment No.	08
Date	08/10/2026
Title	Employee Salary Details Using Interfaces
Week	8
Course Outcome	CO2

1. Objective / Aim

To write a Java program to demonstrate interfaces, implementing interfaces, accessing implementations through interface references, and extending interfaces by calculating monthly salary details for regular and contract employees in a software company.

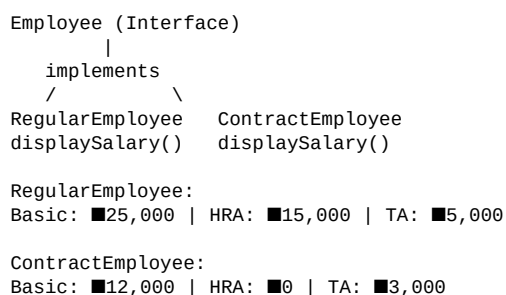
2. System Requirements

Hardware Requirements	Software Requirements
Standard PC or laptop Intel Core i3/i5 processor or equivalent Minimum 4 GB RAM Minimum 500 MB free storage	Windows / Linux / macOS JDK 17 or JDK 21 Eclipse / IntelliJ IDEA / NetBeans / VS Code Command Prompt or Terminal

3. Algorithm / Logic Description

1. Start the program.
2. Define an interface named Employee with a method to display salary details.
3. Create a RegularEmployee class that implements the Employee interface.
4. Set Basic Pay = ₹25,000, HRA = ₹15,000, and TA = ₹5,000 for regular employees.
5. Create a ContractEmployee class that implements the Employee interface.
6. Set Basic Pay = ₹12,000, HRA = ₹0, and TA = ₹3,000 for contract employees.
7. Read the Employee ID from the user.
8. If the ID starts with R, create a regular employee object; if it starts with C, create a contract employee object.
9. Access the implementation through an Employee interface reference.
10. Display the salary details and total monthly salary.
11. Stop the program.

4. UML Class Diagram



5. Complete Java Source Code

```
import java.util.Scanner;

// Defining an interface
interface Employee {
    void displaySalary();
}

// Regular employee implementation
class RegularEmployee implements Employee {
    public void displaySalary() {
        int basicPay = 25000;
        int hra = 15000;
        int ta = 5000;
        int total = basicPay + hra + ta;

        System.out.println("Salary Details:");
        System.out.println("Basic Pay: " + basicPay
            + " HRA: " + hra
            + " T.A: " + ta
            + " Total Amount: " + total);
    }
}

// Contract employee implementation
class ContractEmployee implements Employee {
    public void displaySalary() {
        int basicPay = 12000;
        int hra = 0;
        int ta = 3000;
        int total = basicPay + hra + ta;

        System.out.println("Salary Details:");
        System.out.println("Basic Pay: " + basicPay
            + " HRA: " + hra
            + " T.A: " + ta
            + " Total Amount: " + total);
    }
}

public class EmployeeSalary {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Employee Id: ");
        String empId = sc.nextLine().trim();

        // Access implementation through interface reference
        Employee emp;

        if (empId.matches("(?i)R\\d+")) {
            emp = new RegularEmployee();
            emp.displaySalary();
        } else if (empId.matches("(?i)C\\d+")) {
            emp = new ContractEmployee();
            emp.displaySalary();
        } else {
            System.out.println("Invalid Employee Id");
        }

        sc.close();
    }
}
```

6. Compilation and Execution Commands

```
javac EmployeeSalary.java
java EmployeeSalary
```

7. Output Screen

Test Case 1: Regular Employee

```
Enter Employee Id: R101
Salary Details:
Basic Pay: 25000 HRA: 15000 T.A: 5000 Total Amount: 45000
```

Test Case 2: Contract Employee

```
Enter Employee Id: C101
Salary Details:
Basic Pay: 12000 HRA: 0 T.A: 3000 Total Amount: 15000
```

Test Case 3: Invalid Employee ID

Enter Employee Id: X101
Invalid Employee Id

8. Real-Time Applications

1. Payroll management: Calculate salaries for different categories of employees.
2. Human resource management: Manage regular, contract, and temporary employees.
3. Banking systems: Implement common operations through interfaces.
4. Payment systems: Provide different implementations for payment methods.
5. Enterprise software: Use interfaces to make applications flexible and easier to maintain.

9. Result

Thus, the Java program demonstrating interfaces and accessing their implementations through interface references was successfully implemented. The program calculates and displays monthly salary details for regular and contract employees based on the employee ID.